PROPERTY F IMPLIES WEAK INTEGRALITY FOR PSEUDO-UNITARY BRAIDED FUSION CATEGORIES

Abstract. We prove that every pseudo-unitary braided fusion category with property F is weakly integral. Here property F means that all braid group representations arising from the braiding have finite image. More generally, the conclusion holds for every braided fusion category with property F that has a pseudo-unitary Galois conjugate. This establishes one direction of the Naidu–Rowell property F conjecture for all pseudo-unitary, and hence all unitary, braided fusion categories.

The proof is by contradiction. Using Galois symmetry of the modular data, we construct a non-invertible, self-dual simple object Y in a pseudo-unitary Drinfeld centre, lying in a subcategory with property F. A multi-strand trace inequality, combined with the balancing axiom, forces every simple constituent of $Y \otimes Y$ to have twist 1 or -1. A three-strand Markov identity, together with Galois symmetry, then forces Y to be invertible, giving a contradiction.

1. Introduction

Let $\mathcal{C}$ be a braided fusion category over $\mathbb{C}$. For an object X and an integer $n \geq 2$, the braiding defines a representation

$$\rho_{X,n} : B_n \longrightarrow \mathrm{Aut}(X^{\otimes n}), \qquad \sigma_i \longmapsto \mathrm{id}_X^{\otimes(i-1)} \otimes c_{X,X} \otimes \mathrm{id}_X^{\otimes(n-i-1)}.$$

We write $G_{X,n}$ for its image. The category $\mathcal{C}$ has *property F* if $G_{X,n}$ is finite for every X and n. It is *weakly integral* if $\mathrm{FPdim}(\mathcal{C}) \in \mathbb{Z}$.

Conjecture (Naidu–Rowell [NR])**.** *A braided fusion category has property F if and only if it is weakly integral.*

Thus the conjecture predicts that a single arithmetic condition on the category determines finiteness of its braid group images, for every object and every number of strands.

The implication from weak integrality to property F is known for weakly group-theoretical braided fusion categories [GN, Theorem 7.2], extending the group-theoretical case [ERW]. Etingof, Nikshych and Ostrik asked whether there exists a weakly integral fusion category that is not weakly group-theoretical [ENO2, Question 2]; a negative answer would establish that implication in general. In the other direction, infinite braid group images have been established for several families of quantum-group categories [FLW, LRW]. We prove the following general converse

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under a pseudo-unitarity hypothesis. Recall that a fusion category is *pseudo-unitary* if its global dimension $\dim(\mathcal{C})$ equals its Frobenius–Perron dimension $\mathrm{FPdim}(\mathcal{C})$; unitary fusion categories are pseudo-unitary.

Theorem 1.1. *Let $\mathcal{C}$ be a braided fusion category with a pseudo-unitary Galois conjugate. If $\mathcal{C}$ has property F, then $\mathcal{C}$ is weakly integral. In particular, every pseudo-unitary braided fusion category with property F is weakly integral.*

In particular, the theorem applies to all unitary braided fusion categories. Equivalently, a non-weakly-integral category in this class admits a braid group representation with infinite image. In fact the result holds object by object: every simple object X with $\mathrm{FPdim}(X)^2 \notin \mathbb{Z}$ in such a category already has infinite braid group images (Corollary 7.1), and the modular data determine an explicit number of strands at which infinitude is certified (Corollary 4.3). The trace inequality behind this (Theorem 4.2) needs no pseudo-unitarity, and Section 8 applies it outside the hypothesis of Theorem 1.1.

The proof is by contradiction. Starting from a category with property F that is not weakly integral, we construct a pseudo-unitary modular category $\mathcal{M}$ and a self-dual, non-invertible simple object Y in $\mathcal{M}$. The object Y lies in a fusion subcategory with property F, and it is the image of the unit label under the Galois symmetry of the modular data.

The two main arguments give incompatible conclusions about Y. First, finiteness of its braid group images bounds the sum of the absolute values of certain categorical traces in a Galois-conjugate category. The balancing axiom forces equality in this bound, which implies that every simple constituent of $Y \otimes Y$ has twist 1 or -1. Second, a three-strand trace identity, together with Galois symmetry, shows that an object obtained from the unit label cannot have this twist pattern unless it is invertible. This contradicts the choice of Y.

Section 2 records the transfer properties and Galois identities used in the proof. Section 3 reduces the theorem to a pseudo-unitary Drinfeld centre and locates Y in a subcategory with property F. Sections 4 and 5 establish the trace inequality and its twist consequence. Section 6 proves the Markov identity and the obstruction for non-invertible images of the unit label. The proof of Theorem 1.1 is completed in Section 7.

2. Preliminaries

In the sequel, all fusion categories are over $\mathbb{C}$, and we suppress associativity and unit constraints. We write $\mathbf{1}$ for the tensor unit and N_{XY}^{Z} for the multiplicity of a simple object Z in $X \otimes Y$. A *simple constituent* of a tensor product is a simple object occurring in it (physicists call these *channels*).

2.1. Transfer of property F.

Lemma 2.1 (Direct summands)**.** *Let Y be a direct summand of an object Z in a braided fusion category. Then $G_{Y,n}$ is a quotient of $G_{Z,n}$ for every $n \geq 2$.*

Proof. Choose morphisms $\iota : Y \to Z$ and $\pi : Z \to Y$ with $\pi\iota = \mathrm{id}_Y$. Naturality of the braiding gives

$$\rho_{Y,n}(b) = \pi^{\otimes n}\rho_{Z,n}(b)\iota^{\otimes n} \qquad (b \in B_n).$$

Thus $\ker \rho_{Z,n} \subseteq \ker \rho_{Y,n}$, which gives the asserted quotient. □

Lemma 2.2 (Braided functors and subcategories)**.** *Property F passes to fusion subcategories and along dominant braided tensor functors.*

Proof. The subcategory assertion is immediate. If $F : \mathcal{C} \to \mathcal{D}$ is a braided tensor functor, its tensor structure identifies $\rho_{F(X),n}(b)$ with $F(\rho_{X,n}(b))$. Hence $G_{F(X),n}$ is a quotient of $G_{X,n}$. If F is dominant, every object of $\mathcal{D}$ is a direct summand of some $F(X)$, so Lemma 2.1 applies. □

Lemma 2.3 (Reverse braidings and tensor products)**.** *Let $\mathcal{C}$ and $\mathcal{D}$ be braided fusion categories. The category $\mathcal{C}$ has property F if and only if $\mathcal{C}^{\mathrm{rev}}$ does. Moreover, $\mathcal{C} \boxtimes \mathcal{D}$ has property F if and only if both $\mathcal{C}$ and $\mathcal{D}$ do.*

Proof. The reverse braiding satisfies $c^{\mathrm{rev}}_{X,X} = c^{-1}_{X,X}$, and $\sigma_i \mapsto \sigma_i^{-1}$ defines an automorphism of B_n. For an external tensor product $X \boxtimes Y$, the braid image is a quotient of a subgroup of $G_{X,n} \times G_{Y,n}$. Every object $\bigoplus_i X_i \boxtimes Y_i$ is a direct summand of $(\bigoplus_i X_i) \boxtimes (\bigoplus_i Y_i)$, so Lemma 2.1 proves the forward implication. The converse follows from the braided embeddings of the two factors. □

Lemma 2.4 (Closure)**.** *Let X be an object of a braided fusion category, and let $\langle X \rangle$ be the fusion subcategory it generates. If $G_{X,n}$ is finite for every n, then $\langle X \rangle$ has property F.*

Proof. First let $Z = m\mathbf{1} \oplus X$ with $m \geq 1$. The braid group permutes the direct-sum components of $Z^{\otimes n}$ indexed by sequences of summand labels, and the pure braid group, which has finite index, preserves each of them. Since the braiding with $\mathbf{1}$ is trivial, a pure braid acts on the component with k copies of X through the braid obtained by deleting the unit strands, so through the finite group $G_{X,k}$. Hence $G_{Z,n}$ is finite. Next, $G_{Z^{\otimes r},n}$ is the image of a subgroup of B_{rn} (cabled braids) acting on $Z^{\otimes rn}$, so it is finite too. Finally, every object of $\langle X \rangle$ is a direct summand of some $X^{\otimes k}$ (the dual X^* occurs in a tensor power of X), hence of some $(m\mathbf{1} \oplus X)^{\otimes r}$. Apply Lemma 2.1. □

Lemma 2.5 (Galois invariance)**.** *Property F and weak integrality are invariant under Galois conjugation.*

Proof. By Ocneanu rigidity, a fusion category admits a model over a number field [ENO, Theorem 2.30], and it has only finitely many braidings [ENO, Remark 2.33]. The braidings are therefore the solutions of a zero-dimensional system of polynomial equations over that field, so they are defined over a finite extension of it. Galois conjugation acts entrywise on braid matrices and preserves equality and distinctness

of matrices. It therefore induces an isomorphism between the original and conjugate braid images. Weak integrality depends only on the fusion ring, which Galois conjugation preserves. □

2.2. **Dimensions and the centre of a braided category.** For a spherical fusion category, d_X denotes the categorical dimension of X and Tr the categorical trace. Simple categorical dimensions and their Galois conjugates are real and nonzero [ENO, Theorem 2.3, Remark 2.5 and Corollary 2.10]: the squared norm $|X|^2$ of a simple object is totally positive, and in a spherical category $|X|^2 = d_X^2$. The global dimension is $\dim(\mathcal{C}) = \sum_X d_X^2$, with X ranging over the simple objects. A pseudo-unitary fusion category has a unique spherical structure with $d_X = \mathrm{FPdim}(X)$ for every simple X [ENO, Proposition 8.23]; we always use this structure on a pseudo-unitary category. A *modular category* is a non-degenerate braided fusion category with a spherical structure and its associated ribbon twist.

Let $\mathcal{D}$ be a braided fusion category. Its *Müger centre* $\mathcal{D}'$ is the fusion subcategory of objects X with $c_{Y,X}c_{X,Y} = \mathrm{id}$ for every object Y. There are fully faithful braided tensor functors

$$\iota_+ : \mathcal{D} \longrightarrow \mathcal{Z}(\mathcal{D}), \quad X \longmapsto (X, c_{X,-}),$$
$$\iota_- : \mathcal{D}^{\mathrm{rev}} \longrightarrow \mathcal{Z}(\mathcal{D}), \quad X \longmapsto (X, c_{-,X}^{-1}).$$

Their images, denoted $\mathcal{D}_+$ and $\mathcal{D}_-$, centralise one another. The Drinfeld centre $\mathcal{Z}(\mathcal{D})$ is non-degenerate and satisfies $\mathrm{FPdim}(\mathcal{Z}(\mathcal{D})) = \mathrm{FPdim}(\mathcal{D})^2$. It is modular if $\mathcal{D}$ is spherical [EGNO, Theorem 7.16.6 and Corollaries 8.20.13 and 8.20.14]. It is pseudo-unitary if $\mathcal{D}$ is: indeed $\dim(\mathcal{Z}(\mathcal{D})) = \dim(\mathcal{D})^2$ [EGNO, Proposition 9.3.4], so $\dim(\mathcal{D}) = \mathrm{FPdim}(\mathcal{D})$ implies $\dim(\mathcal{Z}(\mathcal{D})) = \mathrm{FPdim}(\mathcal{Z}(\mathcal{D}))$. We write $\mathcal{A} \vee \mathcal{B}$ for the fusion subcategory generated by two fusion subcategories $\mathcal{A}$ and $\mathcal{B}$.

Lemma 2.6. *Inside $\mathcal{Z}(\mathcal{D})$, the subcategory $\mathcal{D}_+ \vee \mathcal{D}_-$ is the centraliser of $\iota_+(\mathcal{D}')$. If $\mathcal{D}$ has property F, then $\mathcal{D}_+ \vee \mathcal{D}_-$ has property F.*

Proof. The two images centralise one another, so $\mathcal{D}_- \subseteq \mathcal{D}_+'$, where the prime now denotes the centraliser in $\mathcal{Z}(\mathcal{D})$. The dimension formula for centralisers gives

$$\mathrm{FPdim}(\mathcal{D}_+') = \frac{\mathrm{FPdim}(\mathcal{Z}(\mathcal{D}))}{\mathrm{FPdim}(\mathcal{D}_+)} = \mathrm{FPdim}(\mathcal{D}) = \mathrm{FPdim}(\mathcal{D}_-)$$

[DGNO, Theorem 3.14]. Thus $\mathcal{D}_+' = \mathcal{D}_-$, and $\mathcal{D}_+ \cap \mathcal{D}_- = \iota_+(\mathcal{D}')$. Since $\mathcal{D}_-' = \mathcal{D}_+'' = \mathcal{D}_+$, this gives

$$(\mathcal{D}_+ \vee \mathcal{D}_-)' = \mathcal{D}_+' \cap \mathcal{D}_-' = \mathcal{D}_- \cap \mathcal{D}_+ = \iota_+(\mathcal{D}').$$

Taking centralisers, and using the double-centraliser theorem [DGNO, Theorem 3.10(ii)] in the non-degenerate category $\mathcal{Z}(\mathcal{D})$, proves the first assertion.

Since the two images centralise one another, the tensor product functor

$$\mathcal{D} \boxtimes \mathcal{D}^{\mathrm{rev}} \longrightarrow \mathcal{D}_+ \vee \mathcal{D}_-, \qquad X \boxtimes Y \longmapsto \iota_+(X) \otimes \iota_-(Y),$$

is braided and dominant. Lemmas 2.2 and 2.3 prove the second assertion. □

2.3. **Galois symmetry of modular data.** Let $\mathcal{M}$ be modular. We normalise the twist by

$$\mathrm{ptr}_2(c_{A,A}) = \theta_A \,\mathrm{id}_A$$

for a simple object A, where ptr_2 is the right partial trace. Twists are roots of unity [E, Theorem 5.1], so every Galois conjugate of a twist has absolute value one. For simple objects X and A, set

$$\mathrm{Hopf}(X,A) := \mathrm{Tr}_{X\otimes A}(c_{A,X}c_{X,A}) = \sum_Z N_{XA}^Z \, d_Z \, \frac{\theta_Z}{\theta_X\theta_A}. \tag{1}$$

The second equality is the balancing axiom. We normalise the S-matrix by

$$s_{XY} = \frac{\mathrm{Hopf}(X,Y)}{\sqrt{\dim(\mathcal{M})}}, \qquad \chi_Y(X) = \frac{s_{XY}}{s_{\mathbf{1}Y}},$$

using the positive square root of the global dimension. The characters of the fusion ring $K_0(\mathcal{M})$ are precisely the χ_Y. Fix a cyclotomic field K containing the entries of s, the twists, and i [NS, Theorem 7.1], [DLN, Theorem II]; the positive square root of $\dim(\mathcal{M})$ is an admissible normalisation there. Galois automorphisms are taken in $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$; their action on modular data depends only on their restriction to K.

For a Galois automorphism τ, the character $\tau\circ\chi_Y$ is again a fusion-ring character. Thus there is a permutation $\hat{\tau}$ of the simple objects such that $\tau\circ\chi_Y = \chi_{\hat{\tau}(Y)}$.

The permutation $\hat{\tau}$ acts on simple-object labels through their fusion-ring characters. In particular, $\hat{\tau}(\mathbf{1})$ need not be the tensor unit. This action on labels is distinct from passing to the Galois-conjugate category $\mathcal{M}^\tau$, whose tensor unit is still $\mathbf{1}$. Both actions appear below: the conjugate category in Section 4, and the permutation of labels everywhere else.

Proposition 2.7 (Galois symmetry [CG], [DLN, §4.1])**.** *Let $\mathcal{M}$ be a modular category and τ a Galois automorphism. For every simple object X,*

$$\tau(d_X) = \chi_{\hat{\tau}(\mathbf{1})}(X), \tag{2}$$

$$\tau(d_X) = \pm\frac{d_{\hat{\tau}(X)}}{d_{\hat{\tau}(\mathbf{1})}}, \tag{3}$$

$$\tau(\dim(\mathcal{M})) = \frac{\dim(\mathcal{M})}{d^2_{\hat{\tau}(\mathbf{1})}}. \tag{4}$$

Moreover, $\hat{\tau}(\mathbf{1})$ is self-dual, and

$$\widehat{\tau^{-1}} = \hat{\tau}^{-1}. \tag{5}$$

Proof. Equation (2) follows from $\chi_{\mathbf{1}}(X) = d_X$. The signed-permutation formula [DLN, (4.3)], see also [ENO, Theorem 10.1],

$$\tau(s_{XY}) = \varepsilon_\tau(X) s_{\hat{\tau}(X),Y}, \qquad \varepsilon_\tau(X)\in\{1,-1\},$$

together with $d_X = s_{X\mathbf{1}}/s_{\mathbf{11}}$ and $\dim(\mathcal{M}) = s_{\mathbf{11}}^{-2}$, gives (3) and (4). For self-duality, use $\chi_{Y^*} = \overline{\chi_Y}$ and the fact that complex conjugation commutes with τ on K. Since $\chi_{\mathbf{1}}$ is real-valued, so is $\tau\circ\chi_{\mathbf{1}}$, which implies that $\hat{\tau}(\mathbf{1})$ is self-dual. The

inverse-permutation assertion follows by composing the actions of τ and τ^{-1} on characters. □

Lemma 2.8 (Galois column)**.** *If $Y = \hat{\tau}(\mathbf{1})$, then $\mathrm{Hopf}(X, Y) = d_Y\, \tau(d_X)$ for every simple X.*

Proof. By the chosen normalisation and (2),

$$\mathrm{Hopf}(X, Y) = d_Y \frac{s_{XY}}{s_{\mathbf{1}Y}} = d_Y \chi_Y(X) = d_Y \tau(d_X). \qquad \square$$

3. Reduction to a pseudo-unitary Drinfeld centre

We now pass to a modular category, so that the Galois identities of Section 2 are available. The reduction also identifies a subcategory with property F inside this modular category, and places the simple object needed for the trace arguments in that subcategory. The roles are as follows:

- $\mathcal{M} = \mathcal{Z}(\mathcal{C})$ provides modular data and Galois symmetry;
- $\mathcal{B} := \mathcal{C}_+ \vee \mathcal{C}_- \subseteq \mathcal{M}$ provides property F;
- $Y \in \mathcal{B}$ is the object to which both later arguments apply.

The argument never needs property F for the whole Drinfeld centre.

Proposition 3.1. *Let $\mathcal{C}$ be a pseudo-unitary braided fusion category that has property F and is not weakly integral. Then $\mathcal{M} := \mathcal{Z}(\mathcal{C})$ is a pseudo-unitary modular category that is not weakly integral, and $\mathcal{B} = \mathcal{C}_+ \vee \mathcal{C}_-$ has property F.*

Proof. Lemma 2.6 shows that $\mathcal{B}$ has property F, and $\mathcal{M}$ is pseudo-unitary and modular. Recall that $\mathrm{FPdim}(\mathcal{M}) = \mathrm{FPdim}(\mathcal{C})^2$. Since $\mathcal{C}$ is pseudo-unitary, $\mathrm{FPdim}(\mathcal{C}) = \dim(\mathcal{C})$. This number is a totally positive algebraic integer: each of its Galois conjugates is the positive global dimension of a Galois-conjugate fusion category [ENO, Theorem 2.3 and Remark 2.5]. If $\mathrm{FPdim}(\mathcal{C})^2$ were an integer m, then an irrational $\mathrm{FPdim}(\mathcal{C}) = \sqrt{m}$ would have the negative conjugate $-\sqrt{m}$, contradicting total positivity. Thus $\mathrm{FPdim}(\mathcal{C})$ would be rational, hence an integer, contrary to the hypothesis. □

The next lemma locates every Galois image of the unit label. It says that such an object braids trivially with every simple object of integer dimension.

Lemma 3.2 (Integral centraliser)**.** *Let $\mathcal{M}$ be a pseudo-unitary modular category, τ a Galois automorphism, and $Y = \hat{\tau}(\mathbf{1})$. Then Y centralises every simple object z of $\mathcal{M}$ with $d_z \in \mathbb{Z}$. In particular, Y centralises every integral fusion subcategory of $\mathcal{M}$.*

Proof. Since $d_z \in \mathbb{Z}$, Lemma 2.8 gives $\mathrm{Hopf}(z, Y) = d_Y \tau(d_z) = d_Y d_z$. On the other hand, by (1),

$$\mathrm{Hopf}(z, Y) = \sum_W N_{zY}^W\, d_W\, m_W, \qquad m_W := \frac{\theta_W}{\theta_z \theta_Y},$$

where each m_W has absolute value one, each coefficient $N_{zY}^W d_W$ is positive by pseudo-unitarity, and the coefficients sum to $d_z d_Y$. Equality with $d_z d_Y$ therefore forces

$m_W = 1$ for every W. The monodromy $c_{Y,z}c_{z,Y}$ acts on the W-isotypic part of $z \otimes Y$ by m_W, so it is the identity. □

Now let $\mathcal{C}$ be as in Proposition 3.1, and let $\mathcal{M}$ and $\mathcal{B}$ be as there. Since the algebraic integer $\dim(\mathcal{M}) = \mathrm{FPdim}(\mathcal{M})$ is not an integer, it is irrational. Choose a Galois automorphism τ with $\tau(\dim(\mathcal{M})) \neq \dim(\mathcal{M})$, and set

$$Y := \hat{\tau}(\mathbf{1}), \qquad \sigma := \tau^{-1}.$$

By (4), $d_Y^2 = \dim(\mathcal{M})/\tau(\dim(\mathcal{M})) \neq 1$, so Y is non-invertible. By Proposition 2.7, Y is self-dual.

Lemma 3.3. *The object Y belongs to $\mathcal{B}$.*

Proof. By Lemma 2.6, $\mathcal{B}$ is the centraliser of $\iota_+(\mathcal{C}')$. The Müger centre $\mathcal{C}'$ is a symmetric fusion category. By Deligne's theorem [Del, Corollaire 0.8] it is equivalent to $\mathrm{Rep}(G, z)$ for a finite group G and a central element z of order at most two, so the Frobenius–Perron dimension of each simple object is the dimension of a representation of G. Thus $\mathcal{C}'$ is integral. Since ι_+ is a tensor functor, it preserves Frobenius–Perron dimensions, and these agree with the categorical dimensions in the pseudo-unitary category $\mathcal{M}$. So $\iota_+(\mathcal{C}')$ is an integral fusion subcategory of $\mathcal{M}$, and Lemma 3.2 applies. □

We have therefore obtained a non-invertible, self-dual simple object Y in a pseudo-unitary modular category, contained in a subcategory with property F. The next two sections use this last property to constrain the twists of the simple constituents of $Y \otimes Y$.

4. A multi-strand trace inequality

Let $\mathcal{B}$ be a spherical braided fusion category, and let $Y \in \mathcal{B}$ be simple; self-duality is not needed in this section. Fix a Galois automorphism σ. The conjugate category $\mathcal{B}^\sigma$ has the same fusion rules, dimensions $\sigma(d_W)$ and twists $\sigma(\theta_W)$, and by Lemma 2.5 its braid images are isomorphic to those of $\mathcal{B}$. Throughout this section, categorical traces Tr are taken in $\mathcal{B}^\sigma$, and tr denotes the ordinary trace on a finite-dimensional vector space. Put

$$q := \sigma(d_Y) \in \mathbb{R} \setminus \{0\}, \qquad t := |q|.$$

In the Galois-conjugate category, categorical dimensions can have either sign. Consequently the traces u_λ of the monodromy eigenspaces defined below may cancel substantially: their sum is q^2, whereas the sum $V = \sum_\lambda |u_\lambda|$ of their absolute values can be much larger. We bound V using finiteness of the braid images, or more precisely a unitarity property that finiteness implies.

Partition the simple constituents of $Y \otimes Y$ into classes according to their twists in $\mathcal{B}$. For a class λ, let θ_λ be its common twist. In $\mathcal{B}^\sigma$, the monodromy $c_{Y,Y}^2$ acts on this class by $\sigma(\theta_\lambda/\theta_Y^2)$. Let E_λ be the corresponding spectral projection, and define

$$u_\lambda := \mathrm{Tr}(E_\lambda) = \sum_{c \in \lambda} N_{YY}^c \, \sigma(d_c), \qquad S_\pm := \sum_{\pm u_\lambda > 0} |u_\lambda|, \qquad V := \sum_\lambda |u_\lambda| = S_+ + S_-.$$

These numbers are real, and $S_+ - S_- = q^2$. Set $E = \sum_{u_\lambda > 0} E_\lambda$. On $Y^{\otimes n}$, let P_i be E acting on strands $i, i+1$, and put

$$Q_i^+ = P_i, \qquad Q_i^- = 1 - P_i, \qquad w^+ = S_+, \qquad w^- = -S_-.$$

Thus $w^\pm$ is the categorical trace of $Q_i^\pm$ on the two strands where it acts. Each P_i is a polynomial in $\rho_{Y,n}(\sigma_i^2)$.

Lemma 4.1 (Trace factorisation)**.** *For signs $\varepsilon_1, \dots, \varepsilon_{n-1}$, let F be a product in which each factor $Q_i^{\varepsilon_i}$ occurs exactly once, in any order. Then*

$$\operatorname{Tr}(F) = \frac{\prod_{i=1}^{n-1} w^{\varepsilon_i}}{q^{n-2}}.$$

Proof. We induct on n, the case $n = 2$ being the definition of $w^\pm$. The only factor meeting strand n is $Q_{n-1}^{\varepsilon_{n-1}}$. Write $F = F_1 Q_{n-1}^{\varepsilon_{n-1}} F_2$, where F_1 and F_2 act on the first $n-1$ strands. Simplicity of Y and the definition of $w^\pm$ imply

$$\operatorname{ptr}_n(Q_{n-1}^\pm) = \frac{w^\pm}{q} \operatorname{id}_{Y^{\otimes(n-1)}}.$$

The bimodule property of the partial trace therefore gives

$$\operatorname{Tr}(F) = \frac{w^{\varepsilon_{n-1}}}{q} \operatorname{Tr}(F_1 F_2).$$

Apply the induction hypothesis to $F_1 F_2$. □

Theorem 4.2. *Fix $n \geq 2$, and set $M_n := \sum_W N_{Y^{\otimes n}}^W \, |\sigma(d_W)|$. Suppose that each multiplicity space $\operatorname{Hom}(W, Y^{\otimes n})$ in $\mathcal{B}^\sigma$ carries a positive definite inner product for which the adjacent monodromies $\rho_{Y,n}^\sigma(\sigma_i^2)$, $1 \leq i \leq n-1$, are all unitary. This holds, for example, if $G_{Y,n}$ is finite. Then*

$$V^{n-1} \leq t^{n-2} M_n. \tag{6}$$

Proof. Fix n, and write

$$N_W = N_{Y^{\otimes n}}^W, \qquad H_W = \operatorname{Hom}(W, Y^{\otimes n}).$$

If $G_{Y,n}$ is finite, then so is the braid image in $\mathcal{B}^\sigma$, and averaging over it gives an inner product on each H_W that is invariant under the entire braid image; in particular the hypothesis holds. In general, each P_i is a spectral projection of the unitary monodromy operator $\rho_{Y,n}^\sigma(\sigma_i^2)$, so every P_i is an orthogonal projection.

The idea is to follow one unitary operator through two trace calculations. Set

$$J_i := Q_i^+ - Q_i^- = 2P_i - 1, \qquad U := J_1 J_2 \cdots J_{n-1}.$$

The involution J_i reverses the sign of the negative class traces, so its categorical trace on the two strands where it acts is $w^+ - w^- = S_+ + S_- = V$. Averaging gives positive inner products on the multiplicity spaces, while the categorical trace weights the resulting ordinary traces by the possibly negative dimensions $\sigma(d_W)$.

Each J_i is a unitary involution, so U is unitary on every H_W; the J_i need not commute for this. Writing U_W for the action of U on H_W, we have $|\operatorname{tr}(U_W)| \le N_W$, and hence

$$|\operatorname{Tr}(U)| = \Big|\sum_W \sigma(d_W)\operatorname{tr}(U_W)\Big| \le \sum_W |\sigma(d_W)|\, N_W = M_n. \tag{7}$$

On the other hand, expanding each $J_i = Q_i^+ - Q_i^-$ writes U as a signed sum of products in which each $Q_i^\pm$ occurs exactly once. Lemma 4.1 applies to every term, and gives

$$\operatorname{Tr}(U) = \frac{(w^+ - w^-)^{n-1}}{q^{n-2}} = \frac{V^{n-1}}{q^{n-2}}.$$

Taking absolute values and comparing with (7) proves (6). □

The theorem becomes effective once M_n is bounded. For every simple W, $\sigma \circ d$ is a character of the fusion ring, so $\sigma(d_W)$ is an eigenvalue of the fusion matrix of W. By Perron–Frobenius theory its absolute value is at most $\operatorname{FPdim}(W)$. Writing $f = \operatorname{FPdim}(Y)$, this gives

$$M_n \le \sum_W N_{Y^{\otimes n}}^W \operatorname{FPdim}(W) = f^n. \tag{8}$$

Corollary 4.3 (Strand bound)**.** *If $\mathcal{B}$ has property F, then $V \le t \operatorname{FPdim}(Y)$. Conversely, suppose that $r := V/(t \operatorname{FPdim}(Y)) > 1$. Then $G_{Y,n}$ is infinite for every*

$$n \ge n_0 := \left\lfloor \frac{\log(\operatorname{FPdim}(Y)/t)}{\log r} \right\rfloor + 2.$$

Proof. If $G_{Y,n}$ is finite, then (6) and (8) give $V^{n-1} \le t^{n-2} f^n$, that is, $r^{n-1} \le f/t$. Under property F this holds for all n, and letting $n \to \infty$ gives $r \le 1$. If $r > 1$, it fails as soon as $n - 1 > \log(f/t)/\log r$, which holds for every $n \ge n_0$. □

5. Twists of the simple constituents

Return to the categories $\mathcal{C}$, $\mathcal{M}$ and $\mathcal{B}$, the automorphisms σ and $\tau = \sigma^{-1}$, and the object $Y = \hat{\tau}(\mathbf{1})$ constructed in Section 3. Use the twist classes and weights u_λ of Section 4, with $\mathcal{B} = \mathcal{C}_+ \vee \mathcal{C}_-$, and let $t = |\sigma(d_Y)|$.

Lemma 5.1. *The class weights satisfy*

$$\left|\sum_\lambda u_\lambda\, \sigma(\theta_\lambda)\right| = t \operatorname{FPdim}(Y).$$

Proof. Balancing and Lemma 2.8 give

$$\sum_c N_{YY}^c\, d_c\theta_c = \theta_Y^2 \operatorname{Hopf}(Y,Y) = \theta_Y^2 d_Y \tau(d_Y).$$

Apply σ and take absolute values. Since twists are roots of unity, $|\sigma(\theta_Y)| = 1$. Also $\sigma(\tau(d_Y)) = d_Y = \operatorname{FPdim}(Y)$, so the right-hand side has absolute value $t \operatorname{FPdim}(Y)$. □

Proposition 5.2. *Every simple constituent of $Y \otimes Y$ has twist 1 or -1.*

Proof. By Lemma 3.3, Y belongs to the subcategory $\mathcal{B}$, which has property F. Theorem 4.2 and Lemma 5.1 therefore give

$$t\ \mathrm{FPdim}(Y) = \left|\sum_{\lambda} u_\lambda \sigma(\theta_\lambda)\right| \le \sum_{\lambda} |u_\lambda| = V \le t\ \mathrm{FPdim}(Y).$$

Equality holds throughout. Each weight is nonzero: indeed,

$$u_\lambda = \sigma(m_\lambda), \qquad m_\lambda := \sum_{c\in\lambda} N_{YY}^{c}\, d_c > 0,$$

where positivity follows from pseudo-unitarity. Equality in the triangle inequality consequently implies that there is $\omega \in \mathbb{C}$, with $|\omega| = 1$, such that

$$\sigma(\theta_\lambda) = \mathrm{sgn}(u_\lambda)\,\omega \qquad \text{for every class } \lambda.$$

Since Y is self-dual, the unit occurs in $Y \otimes Y$. Its twist is 1, so $\omega \in \{1, -1\}$. Hence every $\sigma(\theta_\lambda)$ belongs to $\{1, -1\}$, and applying σ^{-1} proves the assertion. □

6. A Markov identity and a Galois obstruction

We first establish two trace identities in an arbitrary ribbon fusion category. Let A be simple and write $c = c_{A,A}$. For a simple constituent X of $A \otimes A$, the braiding acts on the multiplicity space by postcomposition,

$$c_X : \mathrm{Hom}(X, A\otimes A) \to \mathrm{Hom}(X, A\otimes A), \qquad c_X(f) = c \circ f,$$

and we set $t_k(X) := \mathrm{tr}(c_X^k)$. Balancing implies that c_X^2 is the scalar $\theta_X \theta_A^{-2}$.

Lemma 6.1 (Kink identity)**.** *One has*

$$\sum_X d_X\, t_1(X) = \mathrm{Tr}_{A\otimes A}(c) = \theta_A d_A.$$

Proof. The first equality follows by decomposing $A\otimes A$ into simple objects with multiplicity spaces. For the second, take the right partial trace: $\mathrm{Tr}(c) = \mathrm{Tr}_A(\mathrm{ptr}_2(c)) = \theta_A d_A$. □

Lemma 6.2 (Markov identity)**.** *One has*

$$\sum_X t_1(X)\ \mathrm{Hopf}(X, A) = \theta_A\, \mathrm{Tr}_{A\otimes A}(c^4) = \theta_A \sum_X d_X\, t_4(X).$$

In a modular category, Lemma 6.2 is a consequence of the self-braiding trace formula of Bantay [Ban], which expresses each $t_1(X)$ in terms of modular data: pairing that formula with the S-matrix column of A gives the lemma. See Giorgetti–Rehren [GR, Proposition 2.3 of the arXiv version] for a proof in unitary modular categories, and Ng–Rowell–Wen [NRW, Theorem 3] for the premodular setting and its relation to generalised Frobenius–Schur indicators. The lemma itself holds in any ribbon fusion category, and we include a short braid-group proof. We compute the trace of one three-strand braid in two ways. Decomposing the first two strands into simple constituents produces the Hopf-link sum on the left. A braid conjugation followed by a partial trace produces $\theta_A\,\mathrm{Tr}(c^4)$ on the right.

Proof. Let $a = \sigma_1$ and $b = \sigma_2$ act on $A^{\otimes 3}$, and put $\mu = ba^2b$. By the hexagon identities, μ is the block monodromy $c_{A,A\otimes A} \circ c_{A\otimes A,A}$: the first two strands, regarded as a single block, travel once around the third strand. On the summand corresponding to $\mathrm{Hom}(X, A \otimes A) \otimes (X \otimes A)$, naturality identifies the actions of a and μ with

$$c_X \otimes \mathrm{id}_{X\otimes A}, \qquad \mathrm{id} \otimes (c_{A,X} c_{X,A}),$$

respectively. Taking categorical traces gives

$$\mathrm{Tr}(a\mu) = \sum_X t_1(X)\ \mathrm{Hopf}(X, A). \tag{9}$$

The braid relation gives an explicit conjugacy. For $w = a\mu = aba^2b$,

$$w = babab, \qquad b^{-1}wb = abab^2 = bab^3, \qquad b^{-2}wb^2 = ab^4.$$

Conjugation by $\Delta = aba$ exchanges a and b, so w is conjugate to ba^4. Categorical trace is invariant under conjugacy. Taking the partial trace over the third strand therefore yields

$$\begin{aligned} \mathrm{Tr}(a\mu) &= \mathrm{Tr}(ba^4) \\ &= \mathrm{Tr}_{A\otimes A}\big((\mathrm{id}_A \otimes \mathrm{ptr}_2(c))c^4\big) \\ &= \theta_A\, \mathrm{Tr}_{A\otimes A}(c^4). \end{aligned}$$

Together with (9) and the decomposition into multiplicity spaces, this proves the identity. □

Theorem 6.3. *Let $\mathcal{M}$ be a modular category, τ a Galois automorphism, and $A := \hat{\tau}(\mathbf{1})$. If every simple constituent of $A \otimes A$ has twist 1 or -1, then $d_A^2 = 1$.*

Recall that A is self-dual by Proposition 2.7, so $A \otimes A = A \otimes A^*$. The idea of the proof is as follows. The twist assumption restricts the eigenvalues of the rescaled braiding to fourth roots of unity. The kink and Markov identities then give two expressions with the same absolute value, one involving the dimensions d_X and the other their Galois conjugates. Comparing them forces d_A^2 to be fixed by τ, and the Galois identity (4) then forces $d_A^2 = 1$. Pseudo-unitarity is not used.

Proof. On the multiplicity space $\mathrm{Hom}(X, A \otimes A)$, balancing gives

$$(\theta_A c_X)^2 = \theta_X\, \mathrm{id} \in \{\mathrm{id}, -\mathrm{id}\}.$$

Thus $\theta_A c_X$ is diagonalisable with eigenvalues in $\{1, -1, i, -i\}$. Let $n_\mu(X)$ be the multiplicity of the eigenvalue μ. Then

$$t_1(X) = \theta_A^{-1} \sum_\mu \mu\, n_\mu(X), \qquad t_4(X) = \theta_A^{-4} N_{AA}^X. \tag{10}$$

Define the real numbers

$$r := \sum_X (n_1(X) - n_{-1}(X)) d_X, \qquad s := \sum_X (n_i(X) - n_{-i}(X)) d_X.$$

Substituting (10) into the kink identity gives the first equation below. Substituting (10) and the Galois-column identity $\mathrm{Hopf}(X, A) = d_A \tau(d_X)$ of Lemma 2.8 into the

Markov identity gives $\theta_A^{-1} d_A(\tau(r)+i\tau(s)) = \theta_A^{-3}\sum_X N_{AA}^X d_X = \theta_A^{-3} d_A^2$, which is the second:

$$\begin{aligned} r + is &= \theta_A^2\, d_A \quad \text{(kink identity)},\\ \tau(r) + i\tau(s) &= \theta_A^{-2}\, d_A \quad \text{(Markov identity)}. \end{aligned} \tag{11}$$

In the Markov identity the dimension weights d_X are replaced by $\tau(d_X)$, while the eigenvalue labels $1, -1, i, -i$ remain fixed. This is why the second equation contains $i\tau(s)$, even when $\tau(i) = -i$.

Dimensions in a spherical fusion category are real, and so are their Galois conjugates by (3). Since r, s, $\tau(r)$ and $\tau(s)$ are therefore real and $|\theta_A| = 1$, taking squared absolute values in (11) gives

$$r^2 + s^2 = d_A^2, \qquad \tau(r)^2 + \tau(s)^2 = d_A^2.$$

Applying τ to the first equality shows that

$$\tau(d_A^2) = d_A^2. \tag{12}$$

Write $D = \dim(\mathcal{M})$ and $\rho = d_A^2 = D/\tau(D)$, by (4). Choose $m \geq 1$ such that τ^m fixes the cyclotomic field K. Since $\tau(\rho) = \rho$ by (12), the product telescopes:

$$\rho^m = \prod_{j=0}^{m-1} \tau^j(\rho) = \prod_{j=0}^{m-1} \frac{\tau^j(D)}{\tau^{j+1}(D)} = \frac{D}{\tau^m(D)} = 1.$$

Since d_A is real and nonzero, $\rho > 0$, so $\rho = 1$. □

Corollary 6.4. *If, in Theorem 6.3, $\mathcal{M}$ is pseudo-unitary, then A is invertible. Equivalently, in a pseudo-unitary modular category, every non-invertible image $\hat{\tau}(\mathbf{1})$ of the unit label has a simple constituent of $\hat{\tau}(\mathbf{1}) \otimes \hat{\tau}(\mathbf{1})$ whose twist is not ± 1.*

Proof. Pseudo-unitarity gives $\mathrm{FPdim}(A) = d_A = 1$, so A is invertible. □

Remark 6.5. The twist hypothesis makes both traces in (10) expressible through integer eigenvalue multiplicities. The Galois-column identity then relates the kink and Markov identities. Pseudo-unitarity enters only in Corollary 6.4, and that step needs genuine extra input: outside the pseudo-unitary setting, categorical dimension ± 1 does not imply invertibility. For example, in $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\sigma}$, with $\sigma(\varphi) = -\varphi^{-1}$, the product A of the two nontrivial simple objects lies in the Galois orbit of the unit and has $d_A = -1$ but $\mathrm{FPdim}(A) = \varphi^2$ [PSYZ]. (This A does not satisfy the twist hypothesis, so it is not a counterexample to a stronger statement.)

The Galois hypothesis on A cannot be omitted from the theorem. For example, let $SU(4)_4$ be the modular category associated with the Lie algebra of $SU(4)$ at level 4, and let adj denote its adjoint object. The fundamental object V satisfies the fundamental–dual fusion rule

$$V \otimes V^* = \mathbf{1} \oplus \mathrm{adj}.$$

The dimensions and twists in [DGPRZ, Table 2] give $\theta_{\mathrm{adj}} = -1$ and $d_V^2 = 4 + 2\sqrt{2}$. Thus all simple constituents of $V \otimes V^*$ have twist ± 1, although V is non-invertible. The object V is not self-dual.

Nor can $\{1, -1\}$ be replaced by the fourth roots of unity. In $SU(2)_6$ the spin-one object $A = X_2$ has $d_A = 1 + \sqrt{2}$, and by the truncated Clebsch–Gordan rule $A \otimes A = \mathbf{1} \oplus X_2 \oplus X_4$, with twists 1, i and $-i$ [BRWZ, §4]. Yet A is a Galois image of the unit, obtained from $\zeta_{16} \mapsto \zeta_{16}^3$ by the sine formula for the S-matrix.

7. Proof of the main result

Proof of Theorem 1.1. Suppose that $\mathcal{C}$ has a pseudo-unitary Galois conjugate and property F, but is not weakly integral. By Lemma 2.5 we may replace $\mathcal{C}$ by that conjugate, so $\mathcal{C}$ is pseudo-unitary. Proposition 3.1 shows that the modular category $\mathcal{M} = \mathcal{Z}(\mathcal{C})$ is pseudo-unitary and not weakly integral, and that $\mathcal{B} = \mathcal{C}_+ \vee \mathcal{C}_-$ has property F. The construction in Section 3 gives a non-invertible self-dual object $Y = \hat{\tau}(\mathbf{1})$ in $\mathcal{B}$. By Proposition 5.2, every simple constituent of $Y \otimes Y$ has twist 1 or -1. Then Corollary 6.4 shows that Y is invertible, a contradiction. □

The theorem also identifies infinite braid images object by object. This answers the question someone studying a particular object will ask, and it addresses the objectwise form of the conjecture [GHR, Conjecture 1.5].

Corollary 7.1. *Let X be a simple object of a braided fusion category, and suppose that the fusion subcategory $\langle X \rangle$ has a pseudo-unitary Galois conjugate. If $\mathrm{FPdim}(X)^2 \notin \mathbb{Z}$, then $G_{X,n}$ is infinite for every sufficiently large n. In particular, this applies to every such simple object of a pseudo-unitary braided fusion category.*

Proof. If every $G_{X,n}$ were finite, then $\langle X \rangle$ would have property F by Lemma 2.4, and so would be weakly integral by Theorem 1.1. In a weakly integral fusion category every simple object has integer squared Frobenius–Perron dimension [ENO, Proposition 8.27], a contradiction. Once G_{X,n_0} is infinite, tensoring with identities embeds it in $G_{X,n}$ for every $n \geq n_0$. □

The word *simple* matters: a direct sum of simple objects in a weakly integral category can have non-integral squared Frobenius–Perron dimension.

The proof is also effective. In the setting of Section 5, with $\mathcal{M}$ pseudo-unitary and $Y = \hat{\tau}(\mathbf{1})$ non-invertible, Corollary 6.4 gives a simple constituent of $Y \otimes Y$ with twist outside $\{1, -1\}$, while the unit is a constituent with twist 1. Since every class weight u_λ is nonzero, the triangle inequality in Lemma 5.1 is strict: $V > t\ \mathrm{FPdim}(Y)$. Corollary 4.3 therefore produces an explicit strand number n_0, computed from the modular data alone, such that $G_{Y,n}$ is infinite for every $n \geq n_0$. In computations one can certify a smaller n by checking $V^{n-1} > t^{n-2} M_n$ directly, with M_n computed from powers of the fusion matrix.

8. Consequences and remaining questions

8.1. **Subcategories.** Property F passes to fusion subcategories, so the theorem also gives the following obstruction without any Galois-conjugacy hypothesis on the ambient category.

Corollary 8.1. *Let $\mathcal{C}$ be a braided fusion category containing a fusion subcategory $\mathcal{A}$ with a pseudo-unitary Galois conjugate. If $\mathcal{A}$ is not weakly integral, then $\mathcal{C}$ does not have property F.*

Proof. The braiding of $\mathcal{C}$ restricts to $\mathcal{A}$. If $\mathcal{C}$ had property F, then so would $\mathcal{A}$, contradicting Theorem 1.1. □

Similarly, if $\mathcal{C}$ is generated by fusion subcategories $\mathcal{A}_i$, each with a pseudo-unitary Galois conjugate (possibly for different automorphisms), and $\mathcal{C}$ has property F, then every $\mathcal{A}_i$ is weakly integral. So is $\mathcal{C}$. Indeed, for fusion subcategories $\mathcal{A}, \mathcal{B}$ of a braided fusion category, $\operatorname{FPdim}(\mathcal{A})\operatorname{FPdim}(\mathcal{B}) = \operatorname{FPdim}(\mathcal{A}\vee\mathcal{B})\operatorname{FPdim}(\mathcal{A}\cap\mathcal{B})$ [DGNO, Lemma 3.38]. If $\mathcal{A}$ and $\mathcal{B}$ are weakly integral, so is $\mathcal{A}\cap\mathcal{B}$, as its simple objects are simple in $\mathcal{A}$; hence $\operatorname{FPdim}(\mathcal{A}\vee\mathcal{B})$ is a rational algebraic integer, so an integer. Induction on the number of subcategories finishes the argument.

8.2. **Beyond the pseudo-unitary hypothesis.** Theorem 4.2 needs neither pseudo-unitarity nor self-duality, so it gives infinite-image certificates outside the hypothesis of Theorem 1.1. Earlier trace and positivity obstructions of a related kind appear in Rowell's work on non-unitarisable ribbon categories [Row04]. For example, let $\mathcal{R} = \mathcal{C}(\mathfrak{so}_5, 3/2)_{\mathrm{ad}}$, the rank-six adjoint subcategory at $q = e^{\pi i/9}$, whose fusion ring has no pseudo-unitary categorification [Sch, Theorem 1]. Label its simple objects $X_0, \dots, X_5$ as in [Sch, Row05], and set $a = 2\cos(\pi/9)$ and $t = 2\cos(4\pi/9)$. Then $Y = X_2$ satisfies

$$Y \otimes Y \cong \mathbf{1} \oplus X_3 \oplus X_5, \qquad Y^{\otimes 3} \cong 3X_2 \oplus X_3 \oplus 2X_4,$$

with categorical dimensions $d_{X_2} = t$, $d_{X_3} = 1$, $d_{X_4} = -1$ and $d_{X_5} = -a$, and the three constituents of $Y \otimes Y$ have distinct twists. Taking $\sigma = \mathrm{id}$ gives $V = 2 + a$ and $M_3 = 3t + 3$. Using $a = 2 - t^2$ and $t^3 = 3t - 1$,

$$V^2 - tM_3 = 16 - 8t^2 - 4t > 0,$$

since $0 < t < \frac{1}{2}$. By Theorem 4.2 with $n = 3$, the image $G_{Y,3}$ is infinite. This is not a new infinite-image result; it illustrates the reach of the trace obstruction.

Removing pseudo-unitarity from Theorem 1.1 needs a new idea. Without it, Lemma 5.1 only gives $\left|\sum_\lambda u_\lambda \sigma(\theta_\lambda)\right| = t\,|d_Y|$, whereas Corollary 4.3 gives $V \le t\operatorname{FPdim}(Y)$. When $|d_Y| < \operatorname{FPdim}(Y)$ the bounds no longer meet, and a twist class with zero total dimension is invisible to the trace calculation.

8.3. **Further trace inequalities.** The proof of Theorem 4.2 extends in two directions. For simple X and Y, group the constituents of $X \otimes Y$ by twist and let $V_\sigma(X, Y)$ be the sum of the absolute values of the class traces in $\mathcal{B}^\sigma$. Using alternating strands $X, Y, X, Y, \ldots$ and the pure braid group, property F gives

$$V_\sigma(X, Y)^2 \leq |\sigma(d_X)\sigma(d_Y)| \operatorname{FPdim}(X) \operatorname{FPdim}(Y).$$

Similarly, using spectral projections of the full twist on $Y^{\otimes k}$ and blocks of k strands that share one strand with their neighbours, property F gives $V_k \leq t \operatorname{FPdim}(Y)^{k-1}$, where V_k is the analogous sum for the constituents of $Y^{\otimes k}$. Both inequalities use only fusion rules, dimensions and twists.

8.4. **The converse direction.** These methods cannot prove the converse. If $\mathcal{C}$ is weakly integral, then $|\sigma(d_W)| = d_W$ for all simple W, every class weight is positive, and $V = d_Y^2$, $t = d_Y$ and $M_n = d_Y^n$. So Theorem 4.2 holds with equality, independently of the braid image. Likewise every Galois image of the unit is invertible, since $\dim(\mathcal{C}) \in \mathbb{Z}$. The implication from weak integrality to property F is known for weakly group-theoretical categories [GN, Theorem 7.2]. For weakly group-theoretical modular categories with their canonical positive spherical structure, Li and Ruan have shown that the Reshetikhin–Turaev mapping class group representations have finite projective image [LR]. In general the implication would follow from a negative answer to [ENO2, Question 2]. The argument here also leaves open the implication from property F to weak integrality for braided fusion categories with no pseudo-unitary Galois conjugate.

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